Mikhail Trunov

Contact

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Summary

AI/ML engineer with a full-stack background. Currently developing a machine-learning classifier for ultrasound images at ASTRAIA, on top of years of building production medical software in Python. I use AI agents daily to design, build and ship products end to end.

Skills

- AI & ML: image classification, medical imaging, TensorFlow, LangChain, MCP
- AI-assisted development: Claude Code, Codex, OpenClaw (daily use)
- Languages: Python (expert), JavaScript, TypeScript, Rust (beginner)
- Web Frameworks: Django, FastAPI, React, Next.js, expo
- DevOps & Cloud: Docker, Ansible, Terraform, CI/CD, GCP
- Testing: pytest, unittest, jest
- Databases: PostgreSQL, SQLite, Redis
- Other Tools: Git, Linux, REST APIs, OpenAPI/Swagger, Telegram bots

Built by me

KinderCafe.app

Smartphone app project for finding child-friendly locations and events in Germany. Tech stack: Expo, Firebase, Cloud Run with Wagtail backend with postgis.

✏ Projects

Python package for monitoring serial (COM) port communication (eximius8/comportsniffer)

Developed a Python tool to intercept and log data exchanged between a software application and a physical COM port device. Primarily created as a vibe-coding project to explore low-level serial communication and

real-time data sniffing.

Hazard Class Calculator Web App (eximius8/ecowag)

Wagtail-based web application and REST API for classifying chemical substance hazards Built a Wagtail application to calculate hazard classes of compound substances in accordance with Russian environmental regulations. Includes PDF report generation using `LaTeX` via the `pylatex` library.

AutoLISP Script for Involute Gear Profile Generation in AutoCAD (eximius8/evolventa)

2008-2012 My first programming project as a student. Developed an AutoLISP script to automate the drawing of involute gear profiles in AutoCAD. The script calculates gear geometry based on user input and generates corresponding drawings. It also exports gear parameters to an Excel spreadsheet for documentation and further analysis.

Experience

Medical Developer – NEXUS / ASTRAIA, Ismaning, Germany

Nov 2021 – Present (5 years)

- Developing a machine-learning classifier for ultrasound images
- Designed and implemented the FMF (Fetal Medicine Foundation) risk algorithm in Python to assess pregnancy complication risks
- Developed a Django-based web application to host and deliver the FMF algorithm
- Built internal tools using Tauri to support the design and development of the main medical software
- Migrated CI/CD pipelines from Jenkins to Bitbucket Pipelines, improving automation and maintainability

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Education

2016 - 2019 **M.Sc. in Advanced Materials and Processes (MAP)** – FAU Erlangen-Nürnberg, Germany

2010 - 2012 **M.Eng. in Materials Science** – Volgograd State Technical University, Russia

2006 - 2010 **B.Eng. in Industrial Engineering** – Volgograd State Technical University, Russia

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Languages

- English (fluent)
- German (intermediate)
- Russian (Native)